

VARUN PATIL

Security & Distributed Systems Researcher

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PROFILE

Research scientist and systems engineer with a track record of translating novel security and distributed systems concepts into production infrastructure used at Internet scale. Architected the data plane of Meta's Private AI inference platform, designed protocols for next-generation Internet architectures, and built widely adopted open-source software.

PROFESSIONAL EXPERIENCE

Meta Platforms, Inc.

Menlo Park, California

Research Scientist (E5) – WhatsApp

Jun. 2025 – present

- Led the development of the secure and transparent data plane of Meta's Private AI Inference platform
 - Productionized **WhatsApp's Incognito AI**, an **industry-first product** enabling truly private AI interactions
 - Designed the end-to-end encryption protocol for the Trusted Execution Environment (TEE) platform
 - Drove a cross-team effort to securely scale the TEE platform for orchestrating complex AI services
 - Implemented the Remote Attestation library underpinning trust establishment across all TEE services
 - Optimized the data plane across 4 stack layers, improving end-to-end P50 latency by 65%
- Served as the production **incident response expert** for all WhatsApp services running on the Private AI platform
- Built manual and automated review frameworks for enforcing data handling practices for sensitive services

Operant Networks

Los Angeles, California

Software Developer (C++) – Security

Jun. 2023 – Jun. 2025

- Implemented a secure TPM-backed key storage solution for a platform underpinning 1.5% US energy generation
- Designed a new testing framework, internal build tooling infra, and 3x faster cross-platform CI/CD pipelines

University of California, Los Angeles

Los Angeles, California

Graduate Student Researcher – Internet Research Laboratory

Apr. 2021 – Jun. 2025

- Conducted research on Named Data Networking (NDN), a future Internet protocol stack. Selected Publications:
 - [1] **Patil**, Desai, Zhang, 2022. *Kua: A Distributed Object Store over Named Data Networking*. ACM ICN '22
 - [2] Moll, **Patil**, Zhang, Pesavento, 2021. *Resilient Brokerless Publish-Subscribe Over NDN*. IEEE MILCOM '21
- Designed the State Vector Sync transport protocol for distributed synchronization in adversarial networks
- Implemented NDN Distance Vector (ndn-dv), an efficient multi-path routing protocol for named data networks

OPEN SOURCE

Memories | 3,800+ GitHub stars

- Efficient and feature-rich FOSS alternative to Google Photos, supporting 5M+ file libraries per user with <50ms latency
- Project creator and lead developer, coordinating contributions from a large user community and 40+ developers

EDUCATION

University of California, Los Angeles

Los Angeles, California

Ph.D. Computer Science (Advisor: Lixia Zhang)

Dec. 2020 – Jun. 2025

M.S. Computer Science

Sep. 2020 – Dec. 2021

Indian Institute of Technology Bombay

Mumbai, India

B.Tech. Mechanical Engineering

Jun. 2016 – Aug. 2020

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, PHP, Go, Rust

Security: Trusted Execution, Attestation, TPM, E2EE

Networking: Routing, Synchronization, Consensus

Infrastructure: CI/CD, Conan, Linux